

Ἀνθέμιος ὁ Τραλλιανός

Anthemius of Tralles on Paradoxical Machines

eul_aid: vdg

Period: Ὀψιμαρχαία [Late Antique](#)
Dialect: Κοινή τεχνική [Technical Koine](#)
Domain: Μαθηματικά [Mathematics](#)
Format: Πραγματεία [Treatise](#)

Anthemius of Tralles was a Byzantine mathematician, engineer, and architect active in the 6th century CE under Emperor Justinian I. A native of Tralles in Asia Minor, he is most renowned for his collaboration with the architect Isidore of Miletus in designing and constructing Constantinople's Hagia Sophia after 532 CE. The historian Procopius records him as the project's master builder and praises his learning, while anecdotes also attest to his mechanical ingenuity, such as using steam and reflected light to play a prank on a neighbor.

His extant work includes the fragmentary treatise *On Burning-Mirrors*, which explores the geometry of parabolic and ellipsoidal mirrors. The *Suda* lexicon credits him with a book on remarkable mechanical devices, and he may have authored a commentary on Apollonius of Perga's *Conics*.

Anthemius holds a pivotal place in the history of applied mathematics and architecture. His geometric expertise was essential to the revolutionary design of the Hagia Sophia, whose massive dome and pendentives defined Byzantine architecture. His writings also helped transmit Hellenistic mathematical knowledge into the Byzantine and later Islamic scientific traditions.

Works (2)

- Μαθηματικὸν Ἀπόσπασμα ἐκ Βοβίου · [Mathematical Fragment from Bobbio](#) · 7 passages
- Περὶ παραδόξων μηχανημάτων · [On Paradoxical Machines](#) · 10 passages